Failure Analysis Report

FAIL-0033 — Connectivity Failure

1. Failure Summary

Failure ID	FAIL-0033
Failure Type	Connectivity Failure
Severity	Critical
Component	Configuration Service
Application	APP-003 — GalaxySearch
Build Version	5.0.8
Timestamp	2024-06-08 17:58:14
Environment	ENV-008 — Production-like
OS Version	One UI 6.0
Network	5G
Resolution	Wont Fix

2. Test Case Reference

Test Case ID	TC-1559
Title	Ensure media streaming synchronizes correctly across devices when the session to...
Execution ID	EXEC-00181
Test Run	RUN-006 — Regression Cycle 6 — Build 5.2.1
Requirement	REQ-0188 — Reliability — Multilingual Support

3. Expected vs Observed Behaviour

Expected Behaviour	The system should complete NFC tap-to-pay within the defined SLA.
Observed Behaviour	The system exhibited connectivity failure when the battery level drops below 10%.

4. Error Details

Error Message:

```
BluetoothException: device DEVICE-518 disconnected during pairing handshake.
```

Stack Trace / Log Excerpt:

```
com.samsung.health.Authentication.execute(line 349)
```

Log Reference	LOG-57259

5. Executed Steps at Time of Failure

1	Disconnect the wearable and wait 30 seconds.	The device reconnects automatically within 10 seconds.

SYNTHETIC DATA — No real Samsung confidential information

2	Re-connect the hub and confirm auto-recovery.	Server returns HTTP 200 with a valid JSON payload.
3	Initiate device pairing by pressing the 'Connect' button.	A confirmation notification appears on the paired device.

6. Root Cause Analysis

Firmware update binary not validated against checksum before flashing; partial write leaves device in boot loop.

7. Linked Defect

Defect ID	BUG-0520
Title	Connectivity Failure in Configuration Service: BluetoothException: device DEVICE...
Severity	Critical
Priority	Low
Status	VERIFIED
Root Cause	Firmware update binary not validated against checksum before flashing; partial w...
Resolution	Added key rotation broadcast to all replica nodes; integration test verifies pro...
Assigned To	ENG-026

8. Historical Similar Failures

HIST-0032	FAIL-0033	Connectivity Failure	4.2.7	2021-06-29	5	Wont Fix
HIST-0188	FAIL-0189	Connectivity Failure	3.5.3	2021-07-03	3	Open
HIST-0279	FAIL-0280	Connectivity Failure	4.3.1	2021-03-13	9	Wont Fix

9. Synthetic Data Notice

This document is entirely synthetic and intended for AI/RAG research only.

SYNTHETIC DATA — No real Samsung confidential information